

Project Proposal: High-Performance Deep Learning Inference Framework for Attention Mechanisms

1. Project Objective

To design and implement a lightweight, high-performance deep learning inference framework from scratch, specifically tailored to execute Attention-based model architectures. This project aims to bridge the gap between algorithmic understanding and hardware-level execution, providing deep insights into CPU/GPU architectures, memory hierarchies, computational graph orchestration, and advanced CUDA optimization techniques.

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3. Detailed Phase Breakdown

Phase 1: CPU Matrix Multiplication (GEMM) Optimization & Architectural Analysis

Goal: Build a foundational understanding of hardware architecture, memory hierarchies, and performance tuning by optimizing a basic matrix multiplication algorithm on the CPU.

- Implementation: Start with a naive $O(N^3)$ nested-loop matrix multiplication.
- Optimization Steps: Progressively apply optimizations such as loop unrolling, memory tiling/blocking, and SIMD vectorization (e.g., AVX2/AVX-512).
- Architectural Deep Dive:
 - Analyze the differences between CPU and GPU architectures.
 - Study the memory hierarchy (Registers, L1/L2/L3 Cache, Main Memory).
 - Analyze how cache hit rates and memory access patterns (e.g., row-major vs. column-major) impact overall throughput.
- Deliverables: A detailed performance log documenting each optimization step. Must include performance profiling data (e.g., using `perf` or VTune), cache miss rate analysis, and the exact percentage of performance improvement achieved at each step.

Phase 2: Implementation of Core Attention Operators

Goal: Implement the mathematical operators required to build a complete Attention block.

- Implementation: Develop forward-pass algorithms for essential operators:
 - Softmax: Ensure numerical stability (e.g., subtracting the maximum value).
 - Layer Normalization: Calculate mean and variance efficiently.

- Activation Functions: Implement GELU (Gaussian Error Linear Unit).
- QKV Projections & MatMul: Query, Key, Value linear transformations and the scaled dot-product operations.
- Deliverables: A library of modular, unit-tested CPU operators. Correctness must be validated against a standard framework like PyTorch or NumPy using a strict error tolerance (e.g., `1e-5`).

Phase 3: Inference Framework Design & Computational Graph Orchestration

Goal: Build the backbone of the inference engine to parse, schedule, and execute deep learning models.

- Graph Representation: Define the neural network structure using a standardized format like JSON or Protobuf.
- Graph Parser: Build a parser to read the serialized file and construct an in-memory Directed Acyclic Graph (DAG).
- Operator Orchestrator (Execution Engine):
 - Implement topological sorting to determine the correct execution order of nodes.
 - *Supplementary Addition:* Implement a basic memory planner to allocate and reuse intermediate tensors (Memory Pooling) to reduce allocation overhead during inference.
- Deliverables: A working CPU-based inference engine capable of loading a configuration file and executing the full Attention block end-to-end.

Phase 4: GPU Acceleration & CUDA Operator Integration

Goal: Transition the computational workload from the CPU to the GPU to leverage massive parallelism.

- CUDA Implementation: Rewrite the operators from Phase 2 (GEMM, Softmax, LayerNorm, GELU) using CUDA C++.
- Framework Integration: Extend the inference framework to support device placement. Implement Host-to-Device (H2D) and Device-to-Host (D2H) memory transfer mechanisms.
- Optimization: Utilize CUDA streams for asynchronous execution and optimize thread block/grid configurations for maximum occupancy.
- Deliverables: A hybrid CPU/GPU inference framework. *Supplementary Addition:* Profile the GPU operators using NVIDIA Nsight Compute and Nsight Systems to identify memory bandwidth bottlenecks and SM utilization.

Phase 4 (Alternative): Advanced CPU Parallelism & Model Quantization

Goal: Maximize multi-core CPU utilization and explore model compression techniques to accelerate inference and reduce memory footprint.

- Multi-threading (Parallelism): Implement a thread pool or use OpenMP to parallelize the execution of operators across multiple CPU cores. Analyze the scaling efficiency (e.g., performance gain from 1 core to 4 cores to 8 cores) and handle thread synchronization overhead.
- Data Quantization (INT8): Introduce Post-Training Quantization (PTQ). Convert the FP32 weights and activations to INT8.
- Implementation: Write an INT8 Matrix Multiplication kernel. Understand the math behind quantization (scale and zero-point) and dequantization.
- Deliverables: A multi-threaded CPU inference engine supporting both FP32 and INT8 execution. A detailed report comparing the latency, memory usage, and accuracy drop between the FP32 and INT8 models.

Phase 5: Advanced Optimization: Flash Attention & Kernel Fusion

Goal: Address the memory bandwidth bottleneck of standard Attention mechanisms by implementing Mega-Kernels (Kernel Fusion).

- Concept: Standard attention requires reading/writing intermediate results (like the $N \times N \times N$ attention matrix) to

GPU High Bandwidth Memory (HBM), which is slow.

- Implementation: Implement a simplified version of Flash Attention. Use SRAM tiling to compute the Softmax and attention weights in a single fused kernel without materializing the intermediate $N \times NN \times N$ matrix in HBM.
- Deliverables: A working Flash Attention CUDA kernel integrated into the framework. A comparative analysis showing the reduction in memory read/writes and the resulting latency reduction/throughput increase compared to the naive GPU implementation from Phase 4.

Phase 5 (Alternative): CPU Operator Fusion & Cache-Aware Attention (CPU "Flash Attention")

Goal: Address memory bandwidth bottlenecks on the CPU by minimizing memory round-trips to the Main Memory (DRAM) and maximizing L1/L2 cache utilization.

- Operator Fusion (Mega-Kernel Equivalent): Instead of executing MatMul, adding Bias, and applying GELU as three separate steps (which requires writing intermediate results to memory three times), write a fused operator that performs all three operations in a single pass within the CPU registers/L1 cache.
- Cache-Aware Attention (CPU Tiling): The core concept of GPU "Flash Attention" (SRAM tiling) applies equally to CPUs. Standard attention materializes the huge $N \times NN \times N$ attention matrix in DRAM.
- Implementation: Implement a block-tiled attention algorithm that computes the Softmax and attention weights in blocks that fit perfectly into the CPU's L1/L2 cache, avoiding the materialization of the full $N \times NN \times N$ matrix.
- Deliverables: A CPU-optimized, memory-efficient Attention block. Profiling results (using tools like `perf` or Valgrind/Cachegrind) demonstrating a significant reduction in Last Level Cache (LLC) misses and memory bandwidth usage compared to the naive implementation in Phase 2.

Phase 6 (Supplementary): End-to-End Validation, Profiling, & Benchmarking

(Note: Added to ensure a complete, production-ready engineering lifecycle)

- Correctness: Create an automated test suite comparing the framework's output tensors against PyTorch outputs for various sequence lengths and batch sizes.
- Benchmarking: Measure end-to-end latency and memory footprint.
- Final Report: Compile all findings, architectural diagrams, performance charts, and profiling screenshots into a final internship presentation.